

MAE 3240 Problem Set # 3 Cover Page

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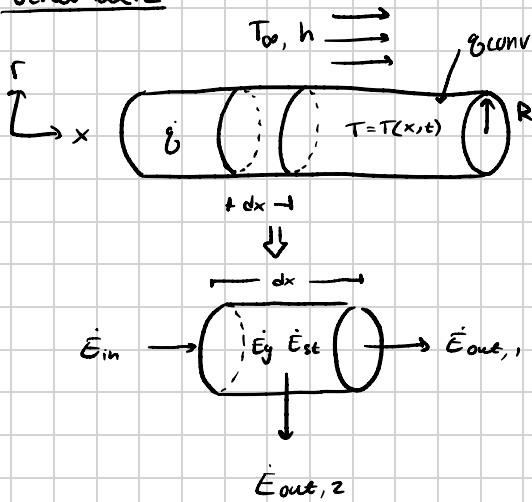
submitted 02-24-26

problem set #3

due 02-24-26

no help

1.

SchematicAssumptions

- Thermal radiation neglected
- Small rod radius
→ uniform temperature along radius
- k , ρ , and c_p are constant

Knowns

- $T = T(x, t)$
- g is uniform

Analysis

a) Energy balance equation for specified control volume

$$\dot{E}_{in} + \dot{E}_g - \dot{E}_{st} = \dot{E}_{out,1} + \dot{E}_{out,2}$$

$$\dot{E}_{in} = \dot{Q}_x(x) = -kA \left. \frac{\partial T}{\partial x} \right|_x$$

$$A_1 = \pi R^2$$

$$A_2 = \pi R^2 dx = 2\pi R = \text{Perimeter, } P$$

$$\dot{E}_{out,1} = \dot{Q}_x(x+dx) = -kA \left. \frac{\partial T}{\partial x} \right|_{x+dx}$$

$$\dot{E}_{out,2} = hA_2(T_s - T_\infty) = h(Pdx)(T_s - T_\infty) = h(2\pi R)(T_s - T_\infty)dx$$

$$\dot{E}_g = \dot{q}V = \dot{q}(A, dx) = \dot{q}(\pi R^2 dx)$$

$$\dot{E}_{st} = \frac{\partial}{\partial t}(\rho c_p (A dx) T) = \rho c_p A, dx \frac{\partial T}{\partial t}$$

$$\dot{E}_{st} = \dot{E}_{in} + \dot{E}_g - \dot{E}_{out,1} - \dot{E}_{out,2}$$

$$-kA \left. \frac{\partial T}{\partial x} \right|_x + \dot{q}(\pi R^2 dx) - \rho c_p A, dx \frac{\partial T}{\partial t} = -kA \left. \frac{\partial T}{\partial x} \right|_{x+dx} + h(2\pi R dx)(T_s - T_\infty)$$

simplifying:

$$kA \left(-\left. \frac{\partial T}{\partial x} \right|_x + \left. \frac{\partial T}{\partial x} \right|_{x+dx} \right) + \dot{q}(\pi R^2 dx) - \rho c_p A, dx \frac{\partial T}{\partial t} = h(2\pi R dx)(T_s - T_\infty)$$

b) Deriving governing equation

let $P = 2\pi R$ and $A = \pi R^2$

$$\dot{E}_{st} = \dot{E}_{in} + \dot{E}_g - \dot{E}_{out,1} - \dot{E}_{out,2}$$

$$\rho c_p A dx \frac{\partial T}{\partial t} = kA \left(-\left. \frac{\partial T}{\partial x} \right|_x + \left. \frac{\partial T}{\partial x} \right|_{x+dx} \right) + \dot{q}(\pi R^2 dx) - h(P dx)(T_s - T_\infty)$$

we want $\frac{\partial T}{\partial t}$:

$$\rho C_p A \frac{\partial T}{\partial t} = k A \left(-\frac{\partial^2 T}{\partial x^2} \right) + \underbrace{\dot{q}}_A (\pi R^2) - \underbrace{h}_{\text{P}} 2 \pi R (T_s - T_\infty)$$

$$\frac{\partial T}{\partial t} = \frac{k A}{\rho C_p A} \left(-\frac{\partial^2 T}{\partial x^2} \right) + \frac{\dot{q}}{\rho C_p A} (\pi R^2) - \frac{h}{\rho C_p} \frac{P}{A} (T_s - T_\infty)$$

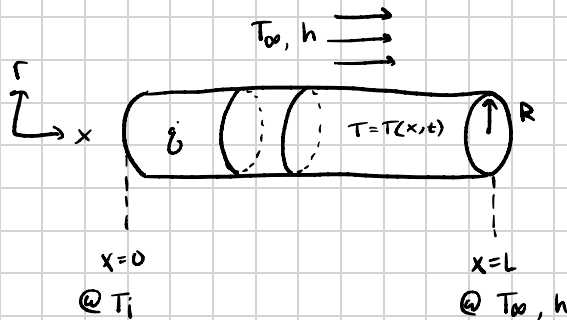
$$\frac{\partial T}{\partial t} = \frac{k}{\rho C_p} \left(-\frac{\partial^2 T}{\partial x^2} \right) + \frac{\dot{q}}{\rho C_p} - \frac{h}{\rho C_p} \cdot \frac{2}{R} (T_s - T_\infty)$$

$$\frac{P}{A} = \frac{2 \pi R}{\pi R^2} = \frac{2}{R}$$

c)

Thermal diffusivity $\alpha = \frac{k}{\rho C_p}$

schematic



knowns

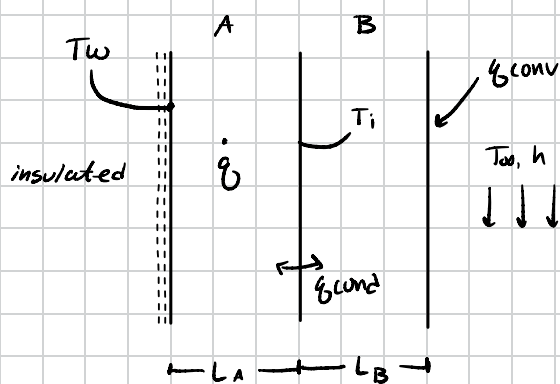
@ $t = 0$ seconds, temp of rod is uniform T_i .

Boundary conditions

$$\begin{aligned} \text{@ } x=0 \\ T(0, t) = T_i \end{aligned}$$

$$\begin{aligned} \text{@ } x=L \\ \text{convective heat transfer} \\ -k A \frac{\partial T}{\partial x} \bigg|_{x=L} = h A (T(L, t) - T_\infty) \end{aligned}$$

2.

SchematicAssumptions

- uniform heat generation ($\dot{q}$)
- left surface of A is thermally insulated
- no heat generation in B
- left surface of B in contact w/ A
- no contact thermal resistance @ boundary
- ideal temperature sensors
- 1D
- steady-state operation

AnalysisKnowns

- $L_A = 30 \text{ cm} = 0.3 \text{ m}$
- $L_B = 10 \text{ cm} = 0.1 \text{ m}$
- $k_B = 30 \text{ W/(m}\cdot\text{K)}$
- $T_\infty = 20^\circ\text{C}$
- $h = 500 \text{ W/(m}^2\cdot\text{K)}$
- $T_w = 115^\circ\text{C}$
- $T_i = 100^\circ\text{C}$

a) Surface temperature of wall B

conduction @ left side and convection @ right side:

$$\dot{q}'' = -k \nabla T$$

$$\dot{q}'' = \frac{k_B}{L_B} (T_i - T_s) = h (T_s - T_\infty)$$

$$\dot{q}'' = \frac{30}{0.1} (100 - T_s) = 500 (T_s - 20)$$

$$\Rightarrow 300(100 - T_s) = 500(T_s - 20)$$

$$30,000 - 300T_s = 500T_s - 10,000$$

$$800T_s = 40,000$$

$$T_s = 50^\circ\text{C}$$

b) Uniform Volumetric heat generation $\dot{q}$ @ s-s

$$\dot{q}'' = \dot{q} L \Rightarrow \dot{q} = \frac{\dot{q}''}{L_A}$$

$$\dot{q}'' = h(T_s - T_\infty) = 500(50 - 20)$$

$$\dot{q}'' = 15,000 \text{ W/m}^2$$

$$\dot{q} = \frac{15,000}{0.3} = 50,000 \text{ W/m}^3$$

c) Thermal conductivity k_A of wall AWall A has heat generation $\dot{q}$ Assuming steady-state generation: $\frac{d^2T}{dx^2} = -\frac{\dot{q}}{k_A}$ @ $x = 0$, $\left. \frac{dT}{dx} \right|_0 = 0$, wall is insulated on the left

$$\text{so } T(x=0) = T_w = 115^\circ\text{C}$$

$$T(x=L_A=0.3\text{m}) = T_i = 100$$

$$T(x) = T_w - \frac{\dot{q}}{2k_A} x^2 \longrightarrow 100 = 115 - \frac{\dot{q}}{2k_A} (0.30)^2, \dot{q} = 5 \times 10^4 \text{ from (b)}$$

$$100 = 115 - \frac{\dot{q}(0.30)^2}{2} \cdot \frac{1}{k_A} \longrightarrow \frac{1}{k_A} = \frac{(115 - 100)2}{\dot{q} - (5 \times 10^4)(0.30)^2} \longrightarrow k_A = \frac{(5 \times 10^4)(0.30)^2}{2(15)} = 150 \text{ W/(m}\cdot\text{K)}$$

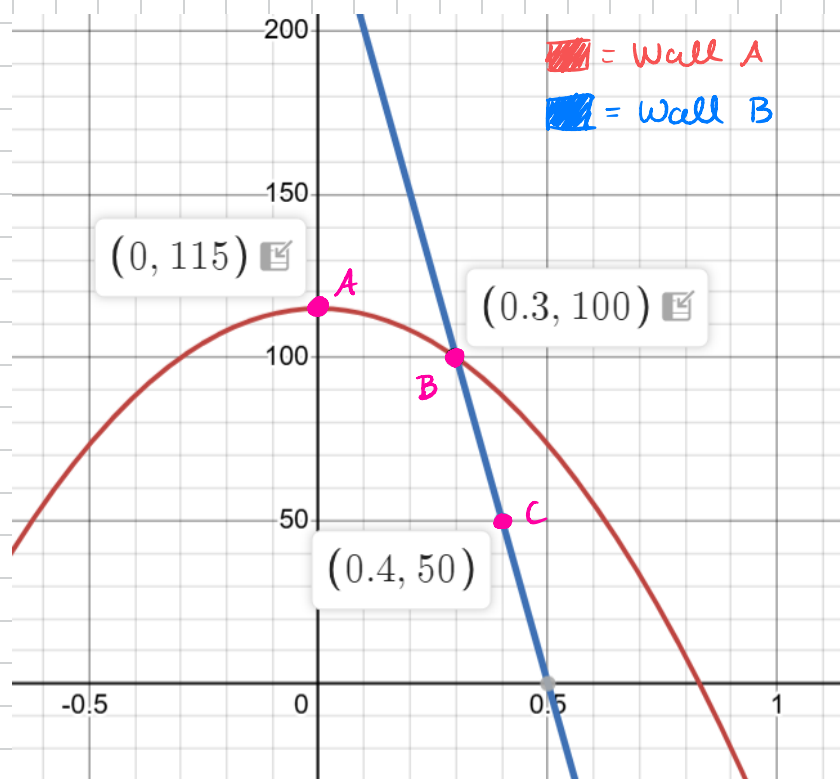
d)

$$T_A(x) = 115 - \frac{\dot{q}}{2k_A} x^2$$

$$T_B(x) = 100 - \frac{\dot{q}}{k_B} (x - L_A)$$

$$T_A(x) = 115 - \frac{(5 \times 10^4)}{2(150)} x^2$$

$$T_B(x) = 100 - \frac{1.5 \times 10^4}{30} (x - 0.30)$$



Important Features

A: Wall A is insulated at $x=0$, so $T_A(x=0) = T_w$ and $\frac{dT_A}{dx} = 0$, shown by

zero slope of $T_A(x)$ at $x=0$

B: Wall A and Wall B have the same temperature at $x = L_A = 0.30 \text{ m}$

C: Wall B has a temperature of 50°C at $x = L_A + L_B = 0.3 \text{ m} + 0.1 \text{ m} = 0.4 \text{ m}$

Slope of T_A

' Slope is parabolic due to uniform heat generation $\dot{q}$ within the wall

Slope of T_B

' Slope is linear because temperature is a linear function of x due to no heat generation within the material

3.

knowns
assumptions
schematic
analysis

Problem summary:

Applying a thermal barrier coating to protect turbine blades from material failure due to high temp gasses produced by the combustor

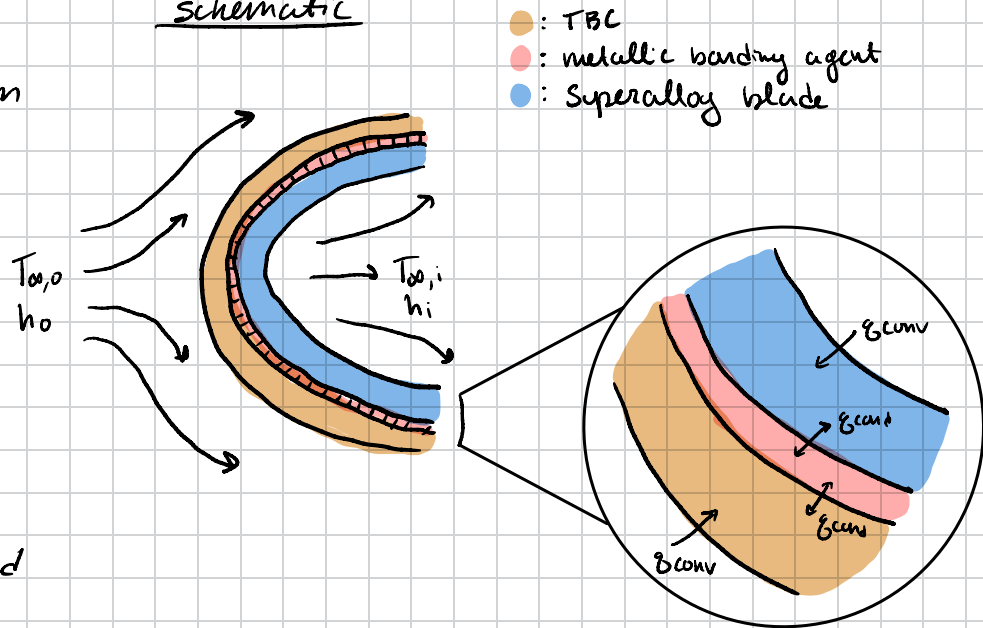
knowns

- $k_B = 25 \text{ W/(m} \cdot \text{K)}$
- $k_{TBC} = 1 \text{ W/(m} \cdot \text{K)}$
- Interfacial thermal resistance
 $R'' = 10^{-4} \text{ m}^2 \cdot \text{K/W}$
- $T_{\infty,0} = 1616 \text{ K}$
- $T_{\infty,i} = 400 \text{ K}$
- $h_o = 1000 \text{ W/(m}^2 \cdot \text{K)}$
- $h_i = 500 \text{ W/(m}^2 \cdot \text{K)}$
- $L_B = 5 \text{ mm} = 0.005 \text{ m}$
- $T_{\max} = 1200 \text{ K}$

Assumptions

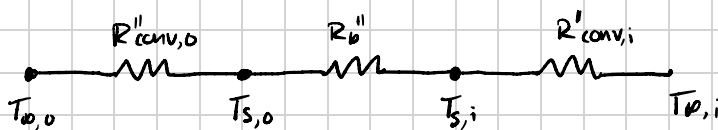
- steady-state operation
- neglecting thermal radiation
- blade approximated as thin wall
- 1D heat conduction
- negligible bonding thicknesses

Schematic



a) TBC is not applied

Thermal resistance network



$$R''_{\text{conv},o} = \frac{1}{h_o} = \frac{1}{1 \times 10^3} = 0.001$$

$$R''_b = \frac{L_b}{k_b} = \frac{0.005}{25} = 0.0002$$

$$R''_{\text{conv},i} = \frac{1}{h_i} = \frac{1}{500} = 0.002$$

$$R''_{\text{tot}} = 0.0032 \text{ m}^2 \cdot \text{K/W}$$

Exterior temp

$$T_{s,0} = T_{\infty,0} - \dot{q}'' R''_{\text{conv},o} \quad \rightarrow \quad T_{s,0} = 1616 - (3.8 \times 10^5)(0.001) = 1236 \text{ K}$$

$$\dot{q}'' = \frac{T_{\infty,0} - T_{\infty,i}}{R''_{\text{tot}}} = \frac{1616 - 400}{0.0032} = 380,000$$

$$1236 > 1200$$

The superalloy blade cannot be kept at a temp under $T_{\max} = 1200 \text{ K}$

b)



$$\frac{L_{TBC}}{k_{TBC}} = \frac{0.0005}{1}, \text{ remaining resistances have the same values as in part (a)}$$

$$R''_{tot} = 0.001 + 0.0005 + 0.0001 + 0.0002 + 0.002 = 0.0038 \text{ m}^2 \text{ K/m}$$

$$q'' = \frac{1216}{3.8 \times 10^{-3}} = 3.2 \times 10^5 \text{ W/m}^2$$

We know that the max temp of the superalloy blade will be on its hot side, facing the hot gases

$$T_{s,0} = 1616 - q''(0.001) = 1616 - 320 = 1296 \text{ K}$$

across the ceramic material

$$\Delta T_{TBC} = q''(0.0005) = 160 \text{ K}$$

$$T_{TBC,in} - \Delta T_{TBC} = 1296 - 160 = 1136 \text{ K}$$

across interfacial region

$$\Delta T_{int} = q''(10^{-4}) = 3.2 \times 10^5 (10^{-4}) = 32 \text{ K}$$

so the temp of the blade is

$$1136 - 32 = \boxed{1104 \text{ K}}$$

$1104 < 1200$, so the temperature can be maintained lower than $T_{max} = 1200 \text{ K}$